

Turning Shopper Data Into Retention & Revenue Strategy

A data analytics case study on customer shopping behavior in apparel & accessories retail — modeled on how data-mature retailers such as Nordstrom operationalize customer analytics to personalize offers, strengthen loyalty, and grow revenue.

Dataset: 3,900 customer transactions across Clothing, Accessories, Footwear & Outerwear

Methodology: Python (pandas) data cleaning → PostgreSQL → SQL business analytics

Deliverables: Cleaned dataset, 10 SQL business insights, this problem statement

Prepared as part of: Customer Shopping Behavior EDA Project

1. Business Context

Apparel and accessories retail is a low-margin, high-competition category where the difference between a thriving brand and a struggling one increasingly comes down to how well a company understands its customers — not just what they buy, but why, how often, and under what conditions (discount, subscription, shipping speed) they choose to buy again.

Nordstrom, the American luxury department store chain, offers a well-documented real-world parallel to the exact dataset this project analyzes. Nordstrom's loyalty program, **The Nordy Club**, has grown to more than 13 million members and is powered by the retailer's internal customer analytics platform, which blends purchase history, discount responsiveness, and shipping preference data to personalize rewards and offers for each shopper. Nordstrom has also invested heavily in improving delivery speed and inventory accuracy, explicitly tying faster, more reliable shipping to higher rates of repeat purchase. Sources: Renascence.io — "How Nordstrom Fosters Customer Loyalty with High-Quality CX"; Ecommerce Fastlane — "Loyalty Program Case Study: Nordstrom"; CX Dive — "Nordstrom hinges CX strategy on personalization and convenience."

Why this matters for this project: The dataset analyzed here contains the same categories of signal Nordstrom uses operationally — subscription status, discount usage, shipping type, review ratings, purchase frequency, and category preference — making it a realistic sandbox for practicing the exact questions a retail analytics team is asked to answer.

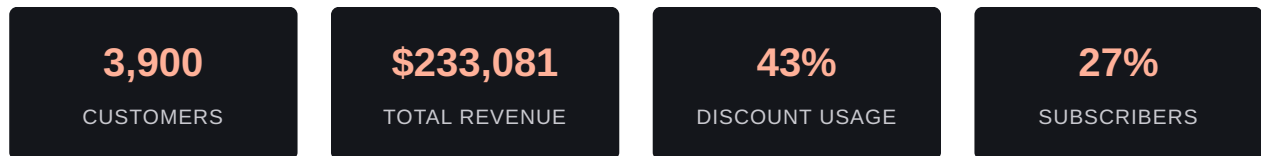
2. The Problem Statement

A mid-sized apparel & accessories retailer has years of raw transactional data sitting in flat files, but no structured way to answer the questions its marketing, merchandising, and retention teams ask every week:

- Which customer segments actually drive revenue — and are we over-indexing on assumptions about who our "best" customer is?
- Is our discounting strategy incremental (driving extra spend) or just a margin giveaway to customers who would have bought anyway?
- Does our subscription program actually correlate with higher spend, or are we crediting it with revenue it didn't create?
- Which products and categories deserve more shelf space, marketing budget, and inventory investment?
- Are our shipping options priced and positioned correctly relative to what customers are actually willing to pay for speed?

Without a clean, query-ready dataset and a repeatable analytics workflow, these questions get answered by anecdote instead of evidence — leading to misallocated marketing spend, mispriced shipping tiers, and loyalty programs that aren't actually tested against the data.

3. Dataset Snapshot



The dataset spans 50 U.S. states, four merchandise categories (Clothing, Accessories, Footwear, Outerwear), six payment methods, and customers aged 18–70. Each record captures the full shopping context of a transaction: product, price, review rating, discount status, subscription status, shipping method, purchase history, and purchase frequency — mirroring the fields a real retail CRM or point-of-sale system would expose to an analytics team.

4. Objectives

Objective	What it resolves
Revenue attribution	Understand how revenue splits across gender, age group, and category to guide budget allocation.
Discount effectiveness	Identify whether discount-driven purchases are still above-average in value, or purely margin erosion.
Loyalty ROI	Test whether subscribed customers actually out-spend non-subscribers, in line with the Nordy Club–style thesis that loyalty membership drives higher lifetime value.
Product strategy	Surface the top-rated and top-selling products, overall and within each category, to guide merchandising decisions.
Shipping economics	Compare spend behavior across shipping tiers to inform free-shipping thresholds and delivery-speed pricing.
Customer segmentation	Classify customers into New, Returning, and Loyal tiers based on purchase history to enable tiered marketing, echoing Nordy Club's tiered rewards structure.

5. Analytical Approach

Phase 1 — Clean & Prepare (Python)

- Audit nulls and impute missing review ratings using category-level means
- Standardize column names and units
- Engineer `age_group` via quartile binning
- Convert qualitative purchase frequency into a numeric `purchase_frequency_days` field
- Remove redundant/duplicate signal columns

Phase 2 — Analyze (SQL / PostgreSQL)

- Load the cleaned dataset into a relational database
- Answer 10 targeted business questions using aggregation, subqueries, CTEs, and window functions
- Export the query-ready table back to CSV for reporting and downstream BI use

6. Key Questions This Case Study Answers

1. **Revenue split** — Total revenue generated by male vs. female customers
2. **Discount behavior** — Customers who used a discount yet still spent above the average purchase amount
3. **Product quality** — Top 5 products by average review rating
4. **Shipping economics** — Average purchase amount, Standard vs. Express shipping
5. **Loyalty ROI** — Average spend and total revenue, subscribers vs. non-subscribers
6. **Discount concentration** — Top 5 products by percentage of discounted purchases
7. **Segmentation** — Customers classified as New, Returning, or Loyal by purchase history
8. **Category leaders** — Top 3 best-selling products within each product category
9. **Repeat-buyer loyalty link** — Whether customers with 5+ prior purchases are more likely to subscribe
10. **Age economics** — Revenue contribution by age group

7. Expected Business Outcomes

Applied at a real retailer, this style of analysis directly supports the same levers Nordstrom uses to grow its loyalty base and per-customer revenue: sharper offer targeting (only discount where it's truly incremental), tiered retention programs that reward the customers actually worth retaining, inventory and marketing weight shifted toward proven top-rated and top-selling products, and a shipping strategy priced around what different customer segments are genuinely willing to pay for speed.

Bottom line: The goal isn't just descriptive reporting — it's building the analytical foundation a retention, merchandising, or marketing team could act on within a single planning cycle.

"91% of consumers are more likely to shop with brands that recognize, remember, and provide relevant offers and recommendations." — Accenture, cited via Flevy's e-commerce fashion retail retention case study.